

$$1) (2x - 3y)dx - (3x - y - 1)dy = 0$$

$$(2x - 3y)dx = (3x - y - 1)dy$$

$$\frac{2x - 3y}{3x - y - 1} = \frac{dy}{dx} \quad \begin{vmatrix} 2 & -3 \\ 3 & -1 \end{vmatrix} = 2 + 9 \neq 0$$

$$\begin{cases} 2x - 3y = 0 \\ y = 3x - 1 \end{cases} \quad \begin{aligned} 2x - 3(3x - 1) &= 0 \\ 2x - 9x + 3 &= 0 \end{aligned}$$

$$-7x = -3$$

$$x = \frac{3}{7}$$

$$x = u + \frac{3}{7} \rightarrow dx = du$$

$$y = v + \frac{2}{7} \rightarrow dy = dv$$

$$y = 3\left(\frac{3}{7}\right) - 1 = \frac{9}{7} - 1 = \frac{2}{7}$$

$$\frac{dv}{du} = \frac{2\left(u + \frac{3}{7}\right) - 3\left(v + \frac{2}{7}\right)}{3\left(u + \frac{3}{7}\right) - \left(v + \frac{2}{7}\right) - 1}$$

$$\frac{dv}{du} = \frac{2u + \frac{6}{7} - 3v - \frac{6}{7}}{3u + \frac{9}{7} - v - \frac{2}{7} - 1}$$

$$\frac{dv}{du} = \frac{2u - 3v}{3u - v}$$

$$\frac{dv}{du} = \frac{2 - 3\frac{v}{u}}{3 - \frac{v}{u}} \quad t = \frac{v}{u} \rightarrow \frac{dv}{du} = t + u \frac{dt}{du}$$

$$t + u \frac{dt}{du} = \frac{2 - 3t}{3 - t}$$

$$u \frac{dt}{du} = \frac{2 - 3t}{3 - t} - t$$

$$u \frac{dt}{du} = \frac{2 - 3t - t(3 - t)}{3 - t}$$

$$u \frac{dt}{du} = \frac{2 - 3t - 3t + t^2}{3 - t}$$

$$u \frac{dt}{du} = \frac{t^2 - 6t + 2}{3 - t}$$

$$\frac{du}{u} = \frac{(3 - t)dt}{t^2 - 6t + 2}$$

$$\begin{aligned} u &= t^2 - 6t + 2 \\ du &= (2t - 6)dt \\ du &= -2(3 - t)dt \end{aligned}$$

$$\ln u = -\frac{1}{2} \ln |t^2 - 6t + 2| + \ln k$$

$$\ln |u| + \frac{1}{2} \ln |t^2 - 6t + 2| = \ln k$$

$$\ln|u^2| + \ln|t^2 - 6t + 2| = \ln k^2 \quad t = \frac{v}{u}$$

$$\ln|u^2 \left(\frac{v}{u}\right)^2 - 6\left(\frac{v}{u}\right)^2 + 2| = \ln k^2$$

$$x = u + \frac{3}{7}$$

$$u = x - \frac{3}{7}$$

$$y = v + \frac{2}{7}$$

$$v^2 - 6vu + 2u^2 = C$$

$$v = y - \frac{2}{7}$$

$$\left(y - \frac{2}{7}\right)^2 - 6\left(y - \frac{2}{7}\right)\left(x - \frac{3}{7}\right) + 2\left(x - \frac{3}{7}\right)^2 = C$$

$$y^2 - \frac{4}{7}y + \frac{4}{49} - (6y - \frac{12}{7})\left(x - \frac{3}{7}\right) + 2\left(x^2 - \frac{6}{7}x + \frac{9}{49}\right) = C$$

$$y^2 - \frac{4}{7}y + \frac{4}{49} - 6xy + \frac{18}{7}y + \frac{12}{7}x - \frac{36}{49} + 2x^2 - \frac{12}{7}x + \frac{18}{49} = C$$

$$y^2 + 2x^2 - 6xy + 2y = C$$

$$2) (x + 2y - 4)dx - (2x + y - 5)dy = 0$$

$$\frac{dy}{dx} = \frac{(x + 2y - 4)}{(2x + y - 5)} \quad \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3$$

$$\begin{cases} x + 2y = 4 \\ 2x + y = 5 \end{cases} \quad \begin{matrix} x = 2 \\ y = 1 \end{matrix}$$

$$\begin{matrix} x = u + 2 & dx = du \\ y = v + 1 & dy = dv \end{matrix}$$

$$\frac{dv}{du} = \frac{(v + 2) + 2(v + 1) - 4}{2(v + 2) + v + 1 - 5}$$

$$\frac{dv}{du} = \frac{v + 2 + 2v + 2 - 4}{2u + 4 + v + 1 - 5}$$

$$\frac{dv}{du} = \frac{v + 2v}{2u + v}$$

$$\frac{dv}{du} = \frac{1 + 2\frac{v}{u}}{2 + \frac{v}{u}}$$

$$t = \frac{v}{u} \quad \frac{dv}{du} = t + u \frac{dt}{du}$$

$$t + u \frac{dt}{du} = \frac{1 + 2t}{2 + t}$$

$$u \frac{dt}{du} = \frac{1 + 2t}{2 + t} - t$$

$$u \frac{dt}{du} = \frac{1 + 2t - t(2 + t)}{2 + t}$$

$$u \frac{dt}{du} = \frac{1 - t^2}{2 + t}$$

$$u \frac{dt}{du} = \frac{1+t^2-2t}{2+t}$$

$$u \frac{dt}{du} = \frac{1-t^2}{2+t}$$

$$\frac{du}{u} = \frac{2+t}{1-t^2} dt$$

$$\frac{du}{u} = \frac{2dt}{1-t^2} + \frac{tdt}{1-t^2}$$

$$\ln|u| = \frac{1}{2} \ln|1+t| - \frac{1}{2} \ln|1-t| - \frac{1}{2} \ln|1-t^2| \quad \left\{ \begin{array}{l} = \frac{A}{1+t} + \frac{B}{1-t} \\ A(1-t) + B(1+t) \end{array} \right.$$

$$u^2 \left| 1 - \left( \frac{v}{u} \right)^2 \right| = C$$

$$u^2 - v^2 = C$$

$$(x-2)^2 - (y-1)^2 = C_1$$

$$x^2 - 2x + 4 - y^2 + 2y + 1 = C_1$$

$$x^2 - y^2 - 2x + 2y = C$$

$$1 = A - At + B + Bt$$

$$1 = A + B$$

$$0 = B - A$$

$$= \frac{1}{2} \left( \frac{1}{1+t} \right) + \frac{1}{2} \left( \frac{1}{1-t} \right)$$

$$1 = \frac{1-t+t+t}{2(1+t)(1-t)}$$

$$\frac{1}{2} \left( \frac{1}{1+t} \right) + \frac{1}{2} \left( \frac{1}{1-t} \right)$$